

Hydra - A Federated Storage System over NDN

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Hydra

- **Sponsored by NSF - Project started in Sept. 2021**
 - https://nsf.gov/awardsearch/showAward?AWD_ID=2126148
- Tennessee Tech
 - Prof. Susmit Shannigrahi
- Clemson
 - Prof. Alex Feltus
- FIU
 - Prof. Alex Afanasyev
- UCLA
 - Prof. Lixia Zhang
- **Goal - Build a federated storage system over NDN**

Justin Presley, Xi Wang, Tianyuan Yu, Proyash Podder, Ishita Debnath, Siqi (Tyler) Liu, Varun Patil



Motivation

- Closing the loop
 - Large science datasets (such as High Energy Physics and Genomics) can benefit from NDN's name-based, location transparent networking
 - Data needs to be available first!
 - *How can we make data available over NDN?*
 - Standalone repositories <- does not scale
 - File system interfaces <- replication and availability is user's responsibility
 - Federated repositories <- this talk
 - We use the genomics community as the driver.



Requirements of the genomics community

- Large storage that can support very large datasets
 - TBs to PBs
- Simple data publication mechanisms
- Provide persistent storage
 - at a file level
- Replicated for availability and supporting geographically distributed users
- Provide transparent data access for the requesters
- Data security and access control



Why Hydra?

- Genomics communities follow a distributed collaborative model. They need:
 - simple and secure publication and access mechanism(s)
 - persistent, replicated data storage
- The current data publication model does not scale
 - **Option 1:** Go through central repositories - selective publication, time consuming
 - **Option 2:** Ad-hoc publication - finding and utilizing data is difficult, lacks replication/resilience
- Storage is available at various institutes but can not be easily utilized. Cloud storage is expensive.
- Data security is currently challenging
 - access control is tied to specific publication systems
 - user credential setups, potentially individual role management and manual role assignment to users
 - data delivery is over individual, separate secured channels.



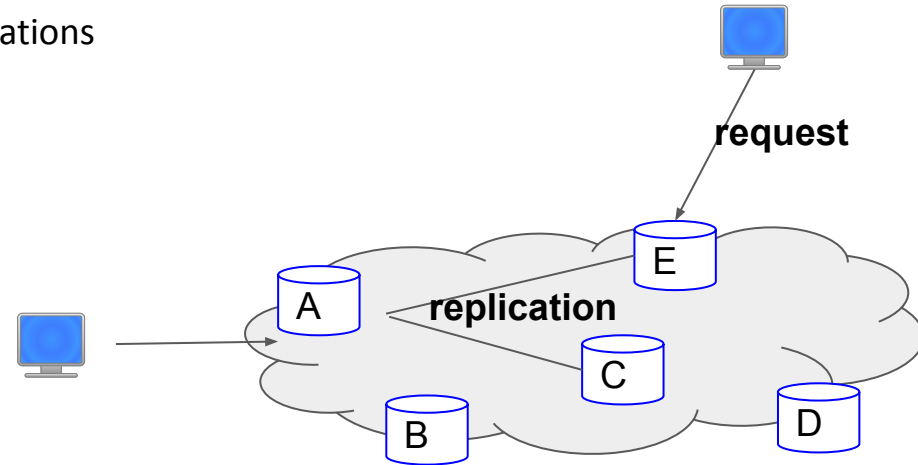
NDN for Distributed repositories

- NDN can support a distributed repository better
 - A loose federation of nodes
 - A uniform namespace - helps with transparent publication, discovery, and retrieval
 - Automated replication - scalability
 - Automated failure recovery
 - Transparent data access using anycast



Hydra - a Federated Storage System Over NDN

- A set of federated repo nodes
 - Utilizes available storage at different organizations
 - Utilizes anycast for data access
- Replicated data files
 - The system automatically decides where to replicate files
 - Automatically maintains the desired degree of replication (default is 3)
- Handle Failures transparently
 - If a node fails, data is automatically replicated to another node

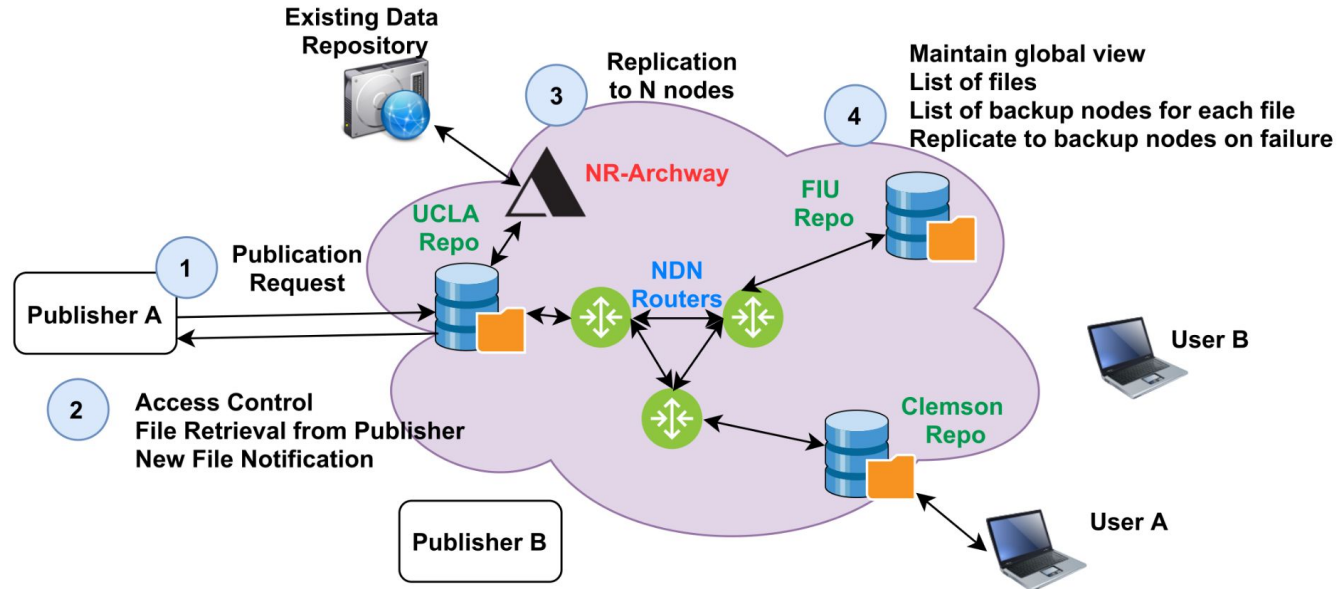


● Secures data using NDN's name based trust schema

● Provides a uniform name based interface for all data

- NDN and non-NDN data

Hydra - a Federated Storage System Over NDN



Design

Assumptions

- **All nodes are equal.** No single point of control
- **Hydra replicates data at the "file" granularity.**
- **Files are immutable.** The repo only handles Insertion and Deletion.
- **Each node will make local decision about which files to replicate (favor).**
- **Repo not responsible for file confidentiality.** If a user wants to store some secret files, they should upload the encrypted ones.
- **No compromised nodes.**



Basic Building Block - Files

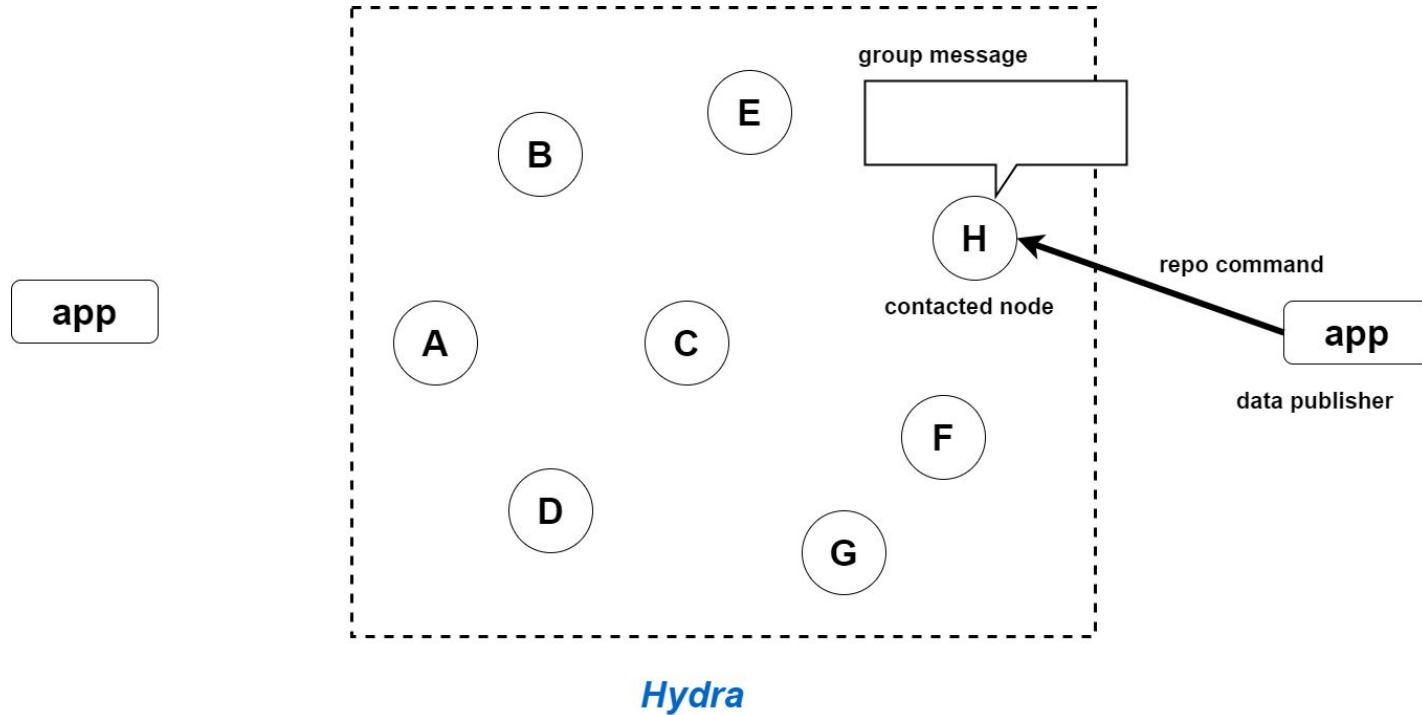
- Users (data publishers & consumers) use Hydra by inserting/deleting/retrieving datasets.
- A "file" is just a BLOB of data identified by a name. For example:
 - A data publisher inserted a "file" whose name is `/BIOLOGY/SRA/SRR5139398`
 - Consumers send an NDN Interest `/hydra/fetch/BIOLOGY/SRA/SRR5139398` .
- We use established naming schemes created by the community



Hydra - Basic Operations

- Data publishers use Hydra by sending Repo Commands (Pub-Sub)
 - Insertion: insert a "file" to Hydra
 - Deletion: delete a "file" from Hydra
 - `/<repo-prefix>/{insert,delete}`
- Data consumers use Hydra by sending NDN Interests directly
 - `/<repo-prefix>/fetch/<file-name>`
- Anyone can send queries to Hydra directly by
 - `/<repo-prefix>/query/QUERY-COMMAND`

Basic Operation



Group Messages & Global View

- **Group Messages.**
 - Hydra use an NDN Sync protocol, the State Vector Sync (SVS), to exchange Group Messages
 - On receiving a global message, a node will update its own Global View
- Each node has a **Global View**
 - Favor - which files a node wants to store
 - Can include more parameters

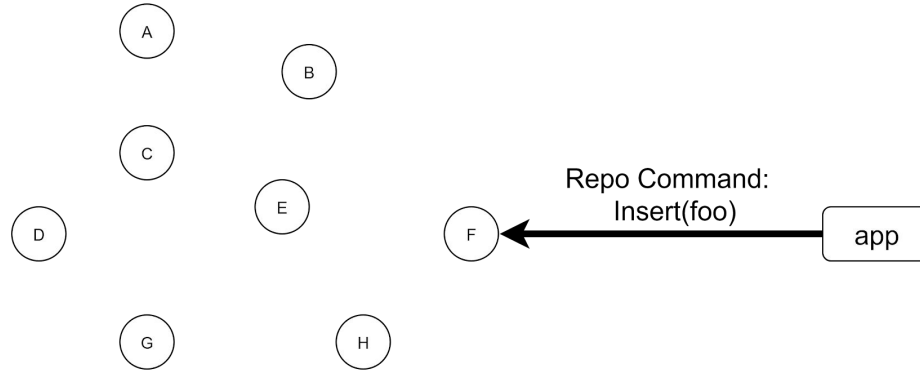
```
[  
  {"node": "D", "favor": 25},  
  {"node": "C", "favor": 20},  
  {"node": "B", "favor": 15},  
  {"node": "G", "favor": 10},  
  {"node": "H", "favor": 5},  
  ....  
]  
  
[]  
  
[]
```

Nodes are Self Organizing

File Insertion

Global View

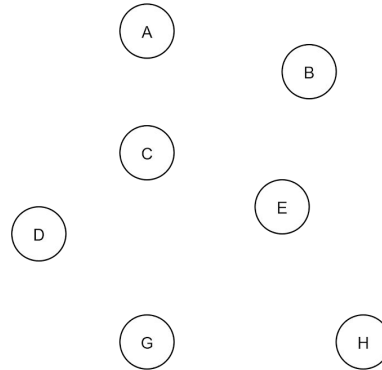
```
[  
  {"node": "D", "favor": 25},  
  {"node": "C", "favor": 20},  
  {"node": "B", "favor": 15},  
  {"node": "G", "favor": 10},  
  {"node": "H", "favor": 5},  
  ...  
]  
  
[]
```



File Insertion

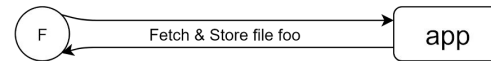
Global View

```
[
  {"node": "D", "favor": 25},
  {"node": "C", "favor": 20},
  {"node": "B", "favor": 15},
  {"node": "G", "favor": 10},
  {"node": "H", "favor": 5},
  ...
]
```



F (the "contacted node")
selects 6 nodes from its
Global View.

F itself might be selected



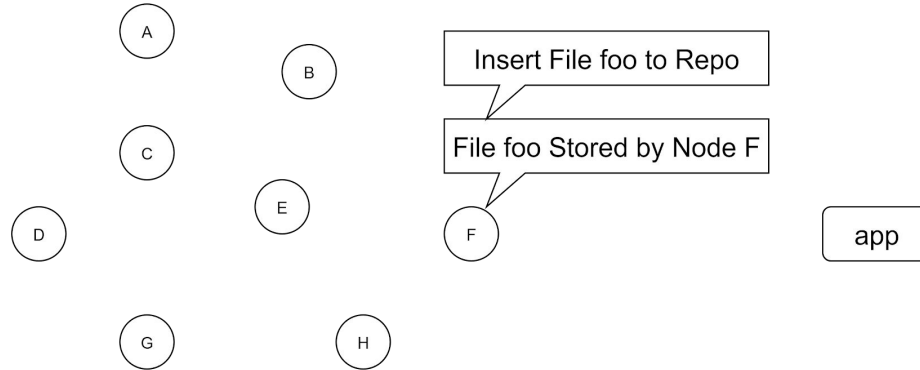
In this case, F selects:
["D", "F", "B", "G", "H", "A"]

File Insertion

Global View

```
[
  {"node": "D", "favor": 25},
  {"node": "C", "favor": 20},
  {"node": "B", "favor": 15},
  {"node": "G", "favor": 10},
  {"node": "H", "favor": 5},
  ...
]
```

```
[{
  "name": "foo",
  "copies": 3,
  ...
  "on": ["F"],
  "backup":
    ["D", "B", "G", "H", "A"]
}]
```

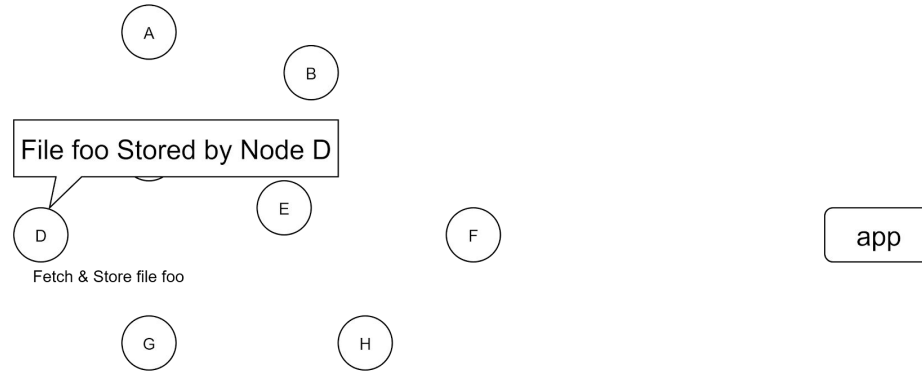


File Insertion

Global View

```
[
  {"node": "D", "favor": 25},
  {"node": "C", "favor": 20},
  {"node": "B", "favor": 15},
  {"node": "G", "favor": 10},
  {"node": "H", "favor": 5},
  ...
]
```

```
[{
  "name": "foo",
  "copies": 3,
  ...
  "on": ["F", "D"],
  "backup":
    ["D", "B", "G", "H", "A"]
}]
```

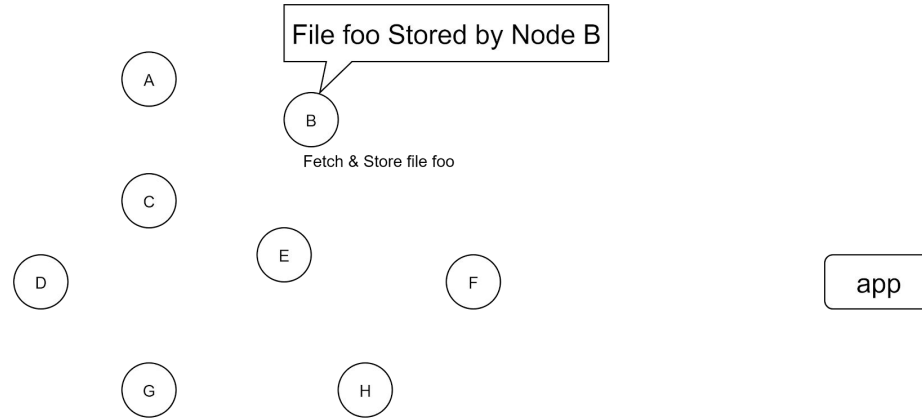


File Insertion

Global View

```
[
  {"node": "D", "favor": 25},
  {"node": "C", "favor": 20},
  {"node": "B", "favor": 15},
  {"node": "G", "favor": 10},
  {"node": "H", "favor": 5},
  ...
]

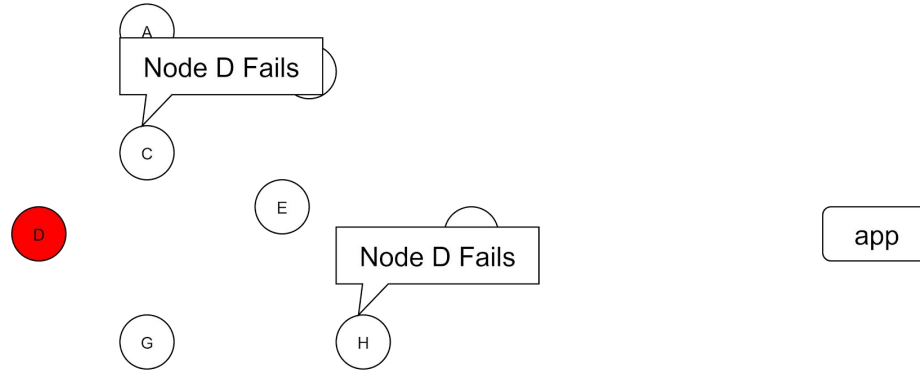
[{
  "name": "foo",
  "copies": 3,
  ...
  "on": ["F", "D", "B"],
  "backup":
    ["B", "G", "H", "A"]
}]
```



Resilience to Node Crashes

Global View

```
[  
  {"node": "D", "favor": 25},  
  {"node": "C", "favor": 20},  
  {"node": "B", "favor": 15},  
  {"node": "G", "favor": 10},  
  {"node": "H", "favor": 5},  
  ...  
]  
  
[{  
  "name": "foo",  
  "copies": 3,  
  ...  
  "on": ["F", "D", "B"],  
  "backup":  
    ["G", "H", "A"]  
}]
```



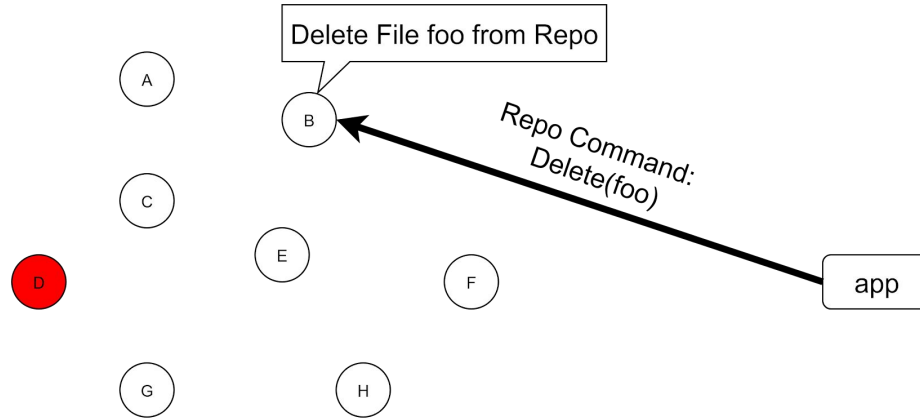
File Deletion

Global View

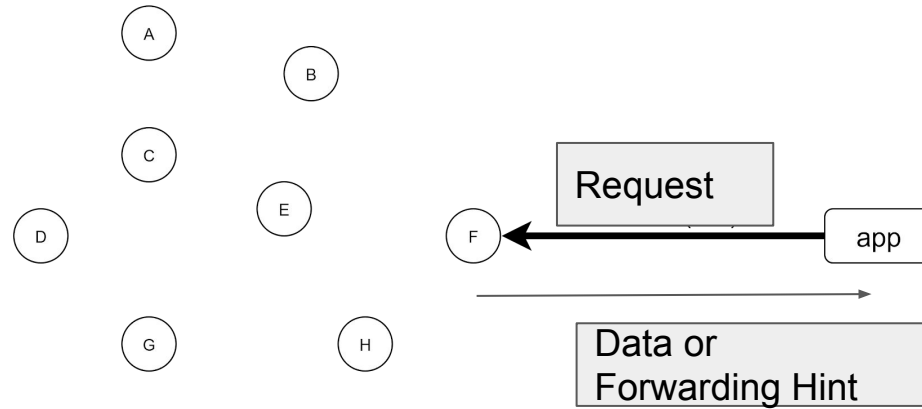
```
[
  {"node": "C", "favor": 20},
  {"node": "B", "favor": 15},
  {"node": "G", "favor": 10},
  {"node": "H", "favor": 5},
  ...
]
```



```
{
  "name": "foo",
  "copies": 3,
  ...
  "on": ["F", "B", "G"],
  "backup":
    ["H", "A", "C"]
}
```

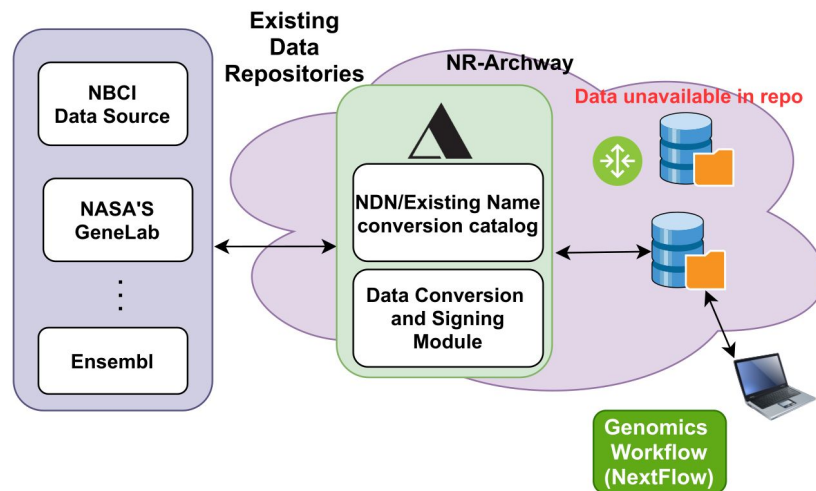


File Fetching



NR-Archway

- Diverse names, metadata standards, and data publication mechanisms
 - wget/curl over HTTP(s), FTP/SFTP, ASPIRA
 - Hardcoded URIs
 - No reusability
 - Findability is poor
- Archway acts as the gateway to all data
 - Converts to NDN as needed
 - Users need not know the publication specifics



Security

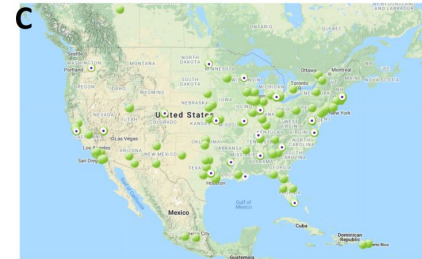
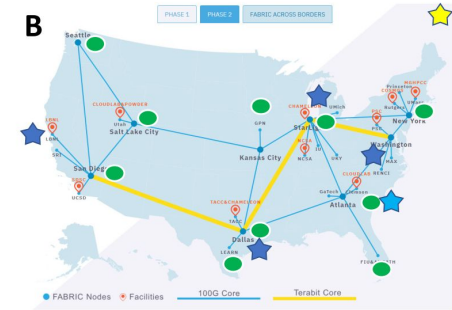
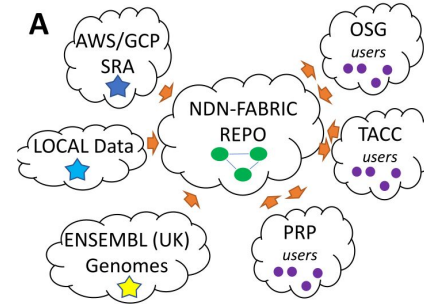
- **Network Operating Centers (NOC):** Serves as a trust anchor
 - Issues certificate for each Hydra node
 - Adds/Deletes Hydra nodes to/from the system
 - Distributes trust policies
- **Hydra Node**
 - Holds certificate from the NOC
 - Executes trust policies fetched from the NOC
- **Clients: Users who insert/delete files**
 - Come from different organizations' namespaces
 - May not share the same trust anchor with Hydra nodes
 - Issue commands that are authenticated by trust policies

Both traditional and attribute based encryption for access control



Testing on FABRIC

- The goal is to deploy Hydra on FABRIC
- Deploy workflows on various cloud/computing platforms
- Large scale experiments with 10-1000 simulated users



- D** Time to Science Variables:
1. Number of Users (10-1000)
 2. User topology
 3. NDN-Repo topology
 4. Retrieval Delay
 5. Network Usage
 6. Caching
 7. Security policy implementation



Conclusions

- Scientific data publication is currently fragmented, findability of data is poor
- Storage exists at various institutes but not federated
 - Can not support large scale genomics
- Security is difficult to configure and implement
- Hydra is a loosely coupled, self-organizing, scalable federated system that aims to address these problems
- Also aims to deliver scalable and easy-to-use security primitives for access control



Questions?

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